GEK-83816

DRIVE SYSTEM INSTRUCTIONS

INSTRUCTIONS For MICROSYNCHRONIZER

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

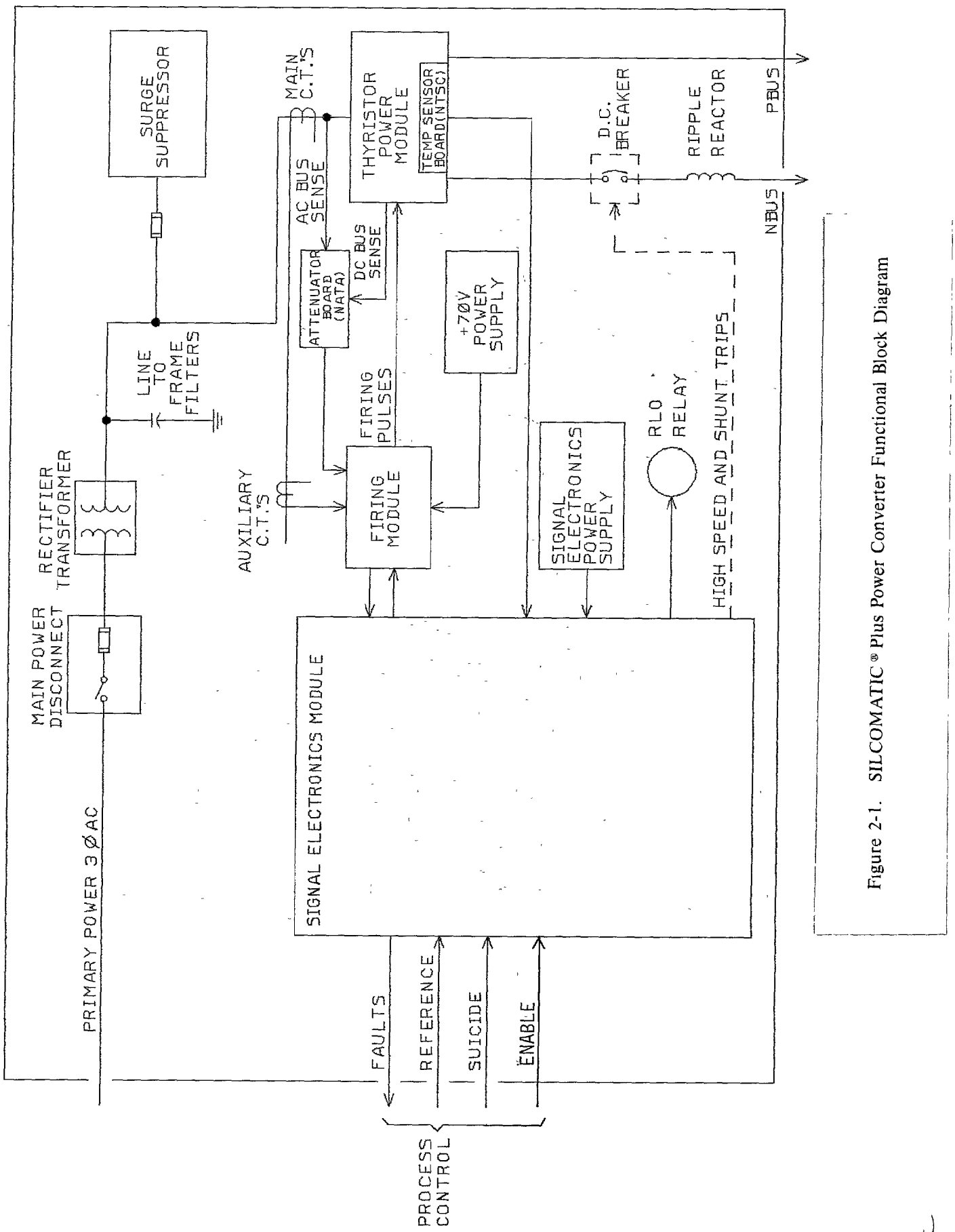


Figure 2-1. SILCOMATIC® Plus Power Converter Functional Block Diagram

PRIMARY POWER 3 Ø AC
MAIN POWER DISCONNECT
RECTIFIER TRANSFORMER
LINE TO FRAME FILTERS
SURGE SUPPRESSOR
SIGNAL ELECTRONICS MODULE
FAULTS
REFERENCE
SYNCS
TIMING
PROCESS CONTROL
MAIN C.T.'S
AC BUS SENSE
THYRISTOR POWER MODULE
TEMP SENSOR BOARD (STATIC)
ATTENUATOR BOARD (MAIN)
DC BUS SENSE
FIRING PULSES
FIRING MODULE
AUXILIARY C.T.'S
+70V SUPPLY POWER
SIGNAL ELECTRONICS SUPPLY POWER
RLO RELAY
HIGH SPEED AND SHUNT TRIPS
BUS
REACTOR
D.C. BREAKER
BUS

GEK-83816

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

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INTRODUCTION

This instruction book describes generator synchronizing hardware known as the microsynchronizer control. It physically consists of two "pieces":

a. a wall-mounted DIRECTOMATIC control isolation module, and

b. a page-mounted DIRECTOMATIC control computer card.

APPLICATION

The microsynchronizer control was designed to work in a General Electric Mark II Speedtronic Turbine Control Panel. It requires power from the P28 and P5 Mark II supplies and specialized pinning for the card.

It may also be applied to a Mark I Turbine Control Panel with the addition of one 5V page-mounted power supply.

For other applications, some repackaging would be necessary.

BACKGROUND

Before initiating synchronization procedures, be sure that all synchronization equipment is functioning properly, and that the phase sequence of the incoming unit corresponds to the existing line phase sequence and the potential transformers are connected correctly to proper phases. Initial synchronization and checkout after performing maintenance to synchronizing equipment should be performed with the breaker racked out.

Generators are synchronized to a power system either manually or automatically.

a. Manual Synchronization. Manual synchronization is accomplished by the station operator who matches the generator terminal voltage phase angle and frequency to that of the power system by using meters, synchroscope and synchronizing lights. When the phase angles, frequencies and voltages are suitably matched, the operator closes the circuit breaker and the generator and system are connected.

b. Automatic synchronization. In automatic synchronization, the same function is accomplished by a synchronizer (speedmatcher/phase angle matcher) with no operator participation. During startup of the turbine-generator, the gas turbine control system brings the generator to approximately 100% speed. The Microsynchronizer control adjusts the speed of the generator and closes the circuit breaker when the frequency and phase of the generator are within acceptable limits of the system bus. It is important that the frequencies and phase angle of the generator voltage and the system are closely matched when the breaker closes. If they are not closely matched, large shaft torques and generator currents may result, causing damage to the turbine and generator.

Gas turbine controls have used automatic synchronization for many years. Experience, particularly in places where system frequencies vary and blackouts happen more often, has led to the conclusion that a faster, more reliable synchronizing system with no adjustments would result in fewer machines damaged by out-of-phase closures. These malsynchronizations are often a result of misadjusted synchronizers or manual breaker closures. There was a need for a fast, reliable device that minimized adjustments.

To minimize the equipment failure and malsynchronization of the turbine driven generator, the Microsynchronizer control was developed to meet these needs for a superior synchronizer.

Figure 1 is a typical one-line diagram of a gas turbine-driven generator. The breaker is the three pole switch used to connect and disconnect the generator to the power system.

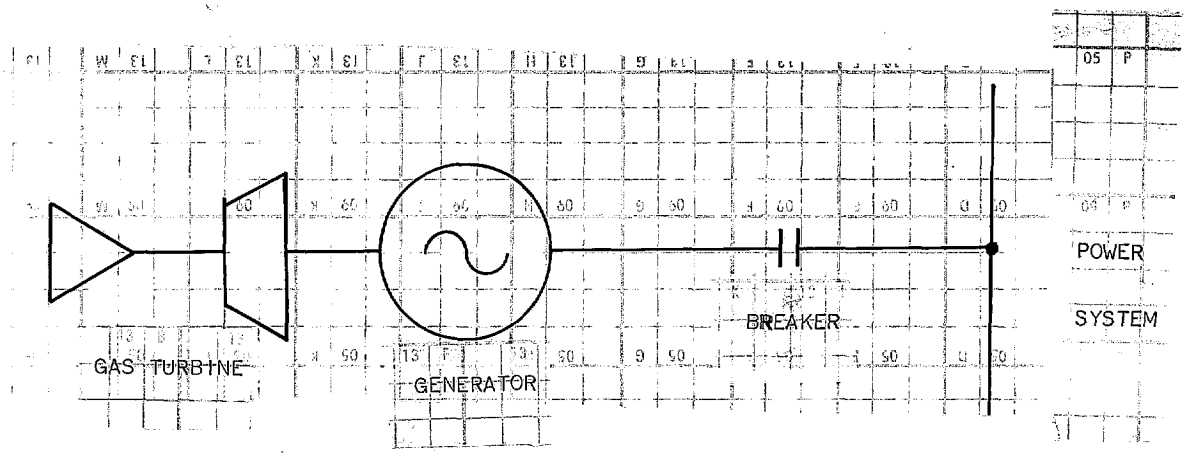


Figure 1
Gas Turbine One-Line Diagram

FUNCTIONAL DESCRIPTION

Figure 2 is a block diagram of the microsynchronizer showing its inputs and outputs. The synchronizer consists of a 7" X 7" X 5" isolation module that mounts inside of the gas turbine control panel and a 4" X 5.4" computer card that is inserted into the turbine control panel card rack. The computer card requires a 5.0 VDC power supply and the isolation module requires a 5.0 VDC and a 28 VDC power supply.

The power line voltage and generator voltage are input to the isolation module. The isolation module conditions these signals and outputs TTL-compatible square waves to the computer card. The isolation module also receives the synchronize command from the computer card and provides a relay contact output to the breaker. This relay applies voltage across the breaker's closing coil.

The computer card contains an INTEL 8748 microcomputer and an INTEL 8253 timer chip. The INTEL 8748 has 1024 bytes of EPROM (Eraseable Programmable Read Only Memory) and 64 bytes of RAM (data memory). When the enable and breaker open inputs are "true", the synchronizer proceeds to synchronize the generator to the power system. Raise or lower speed commands are issued to the gas turbine control system as required and the synchronize command is issued to the isolation module at the correct moment.

The microsynchronizer control performs two primary functions: speed matching and phase angle matching.

The speed matching function is active whenever the "Enable" and "Breaker Open" signals are true. The objective is to bring the turbine generator to a frequency just above the system frequency. The frequency difference between the turbine generator and the line is called slip. The synchronizer drives the turbine into a band of positive slips between 0.09 Hz and 0.15 Hz. When above 0.15 Hz slip, a continuous lower signal is applied to the governor. Below 0.09 Hz, a continuous raise signal is applied to the governor. When between 0.09 and 0.15 Hz slip, neither a raise nor a lower is issued.

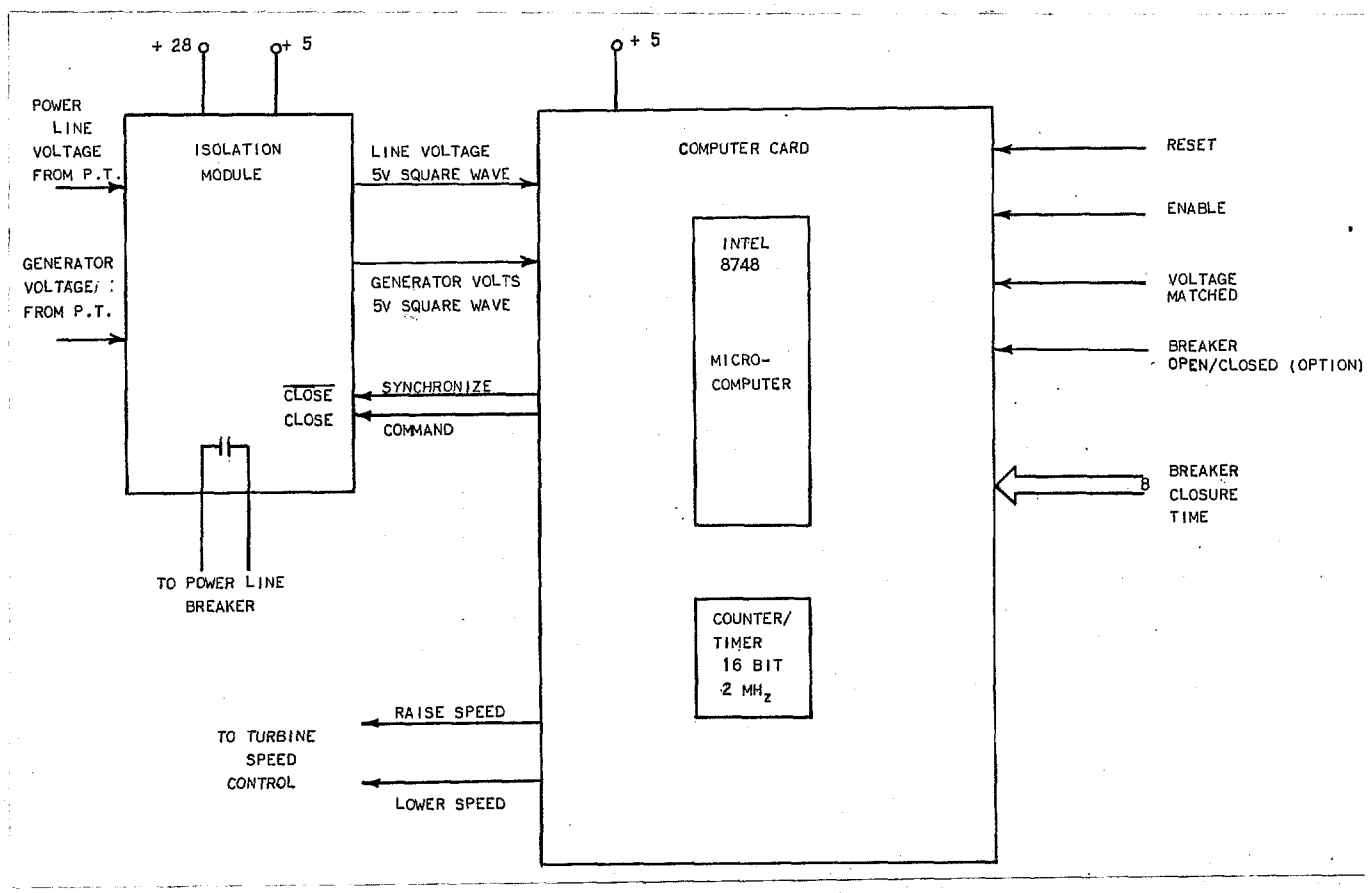


Figure 2
Microsynchronizer Block Diagram

The synchronizing function is activated whenever the "Enable", "Breaker Open", and "voltage matched" signals are true. This function issues a close command to the generator breaker, such that when the breaker closes, the power system and generator voltage are in phase. When this condition exists, the power exchange across the breaker is at a minimum value, and the mechanical stress transients resulting from synchronization are minimized.

Figure 3 is a plot on the phase-slip plane that shows the allowable region for breaker contact closure. The most desired condition for synchronization is the center line of the inner diamond shaped region shown. The outer diamond is the region in which the phase and frequency difference of the generator and the power system must lie before the synchronizer will calculate the optimum time to issue the breaker closure command. The synchronizer will issue the closure command such that the breaker will close when the generator's trajectory reaches the point "closest" to the center line of the diamond shaped target if that point on the trajectory lies within this inner diamond. The target area (inner diamond) was selected because it results in closures that cause no damage to the turbine-generator and gives fast synchronization while providing for slightly positive power generation at closure.

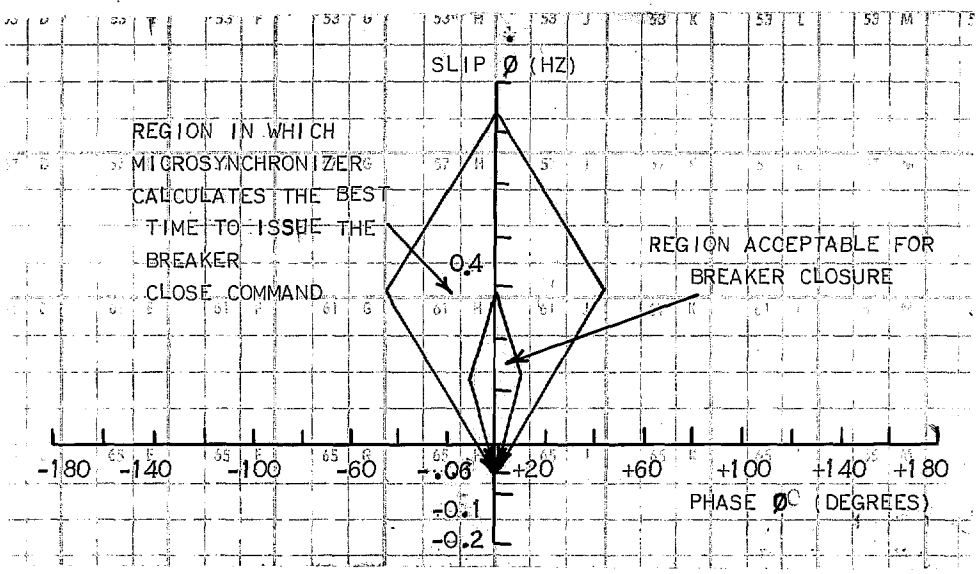


Figure 3
Regions of Calculation and Breaker Closing in the Phase Slip Plane

Figure 3

Figure 4 shows two typical trajectories of the generator during synchronization. For the examples shown, the close command will be issued so that breaker closure will occur at the point indicated. Figure 5 is a more detailed plot of the trajectories in Figure 4 for the region near the target. Since different kinds of breakers have different closing times, the synchronizer must know the closing time of the breaker in use in order to supply the close signal before the generator's trajectory is reached. In order to determine when to issue the synchronize command, the microsynchronizer uses the breaker close time T to predict the angle:

$$\emptyset_F = \emptyset_C + T\dot{\emptyset}_C + .5T^2\ddot{\emptyset}_C \qquad (1)$$

where $\emptyset_C$ is the current phase angle (generator-power system phase difference), $\dot{\emptyset}_C$ is the current slip (generator-power system frequency difference), $\ddot{\emptyset}_C$ is the current acceleration, and $\emptyset_F$ is the predicted value for the phase angle T seconds in the future. The slip value:

$$\dot{\emptyset}_F = \dot{\emptyset}_C + T\ddot{\emptyset}_C \qquad (2)$$

where $\dot{\emptyset}_F$ is the predicted value for slip T seconds in the future, is also calculated. When the values of predicted phase $\emptyset_F$ and predicted slip $\dot{\emptyset}_F$ are determined to be that of the point along the generator's trajectory, that will be closest to the center of the diamond shaped target, the close command is issued.

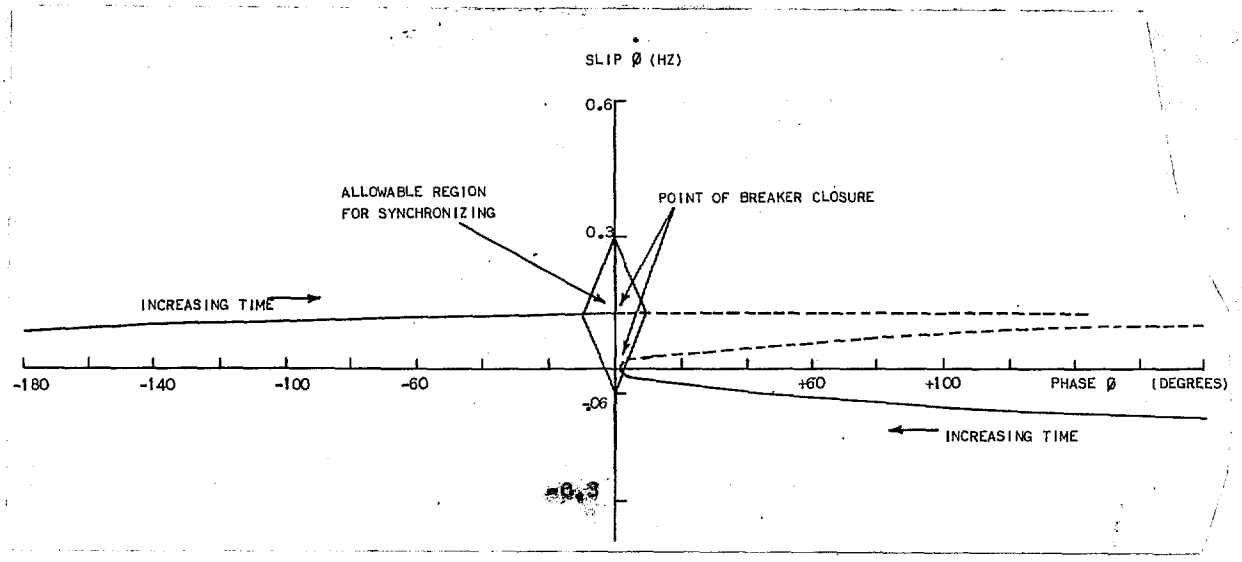


Figure 4
Typical Trajectories During Synchronizing

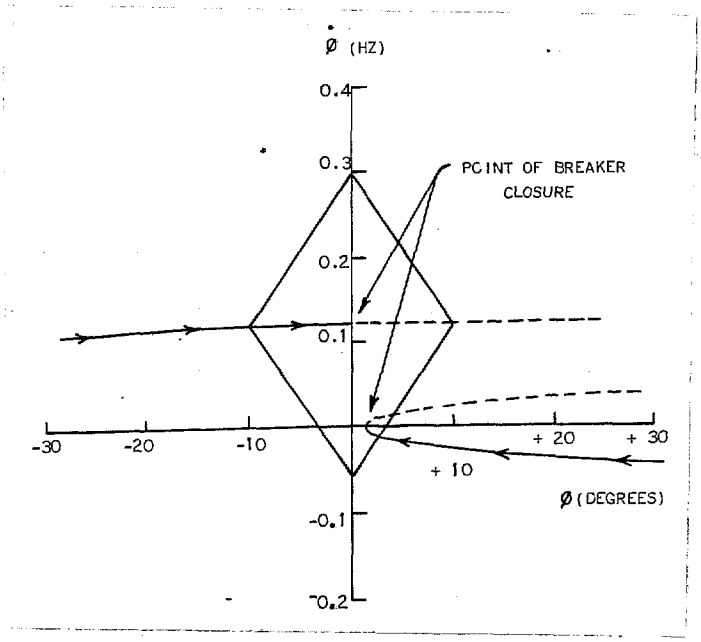


Figure 5
Point of Breaker Closure - Two Typical Trajectories

FEATURES

The Microsynchronizer control includes many features to minimize the probability of closing out of phase. One reason for bad closures on previous systems is misadjustment. The Microsynchronizer control avoids this problem by not having any adjustments.

It must, however, know the breaker close time in order to make the required calculations. For two reasons, this is best supplied from an external input. First, if the card is changed, there will not be any mistake in adjusting for breaker closing time. Second, some applications require using the synchronizer to close two breakers with different close times. The synchronizer is programmed by four pins which represent four bits, giving sixteen states. These sixteen states are interpreted as close times from 5 to 20 cycles where 1 cycle is the period of the power system voltage.

Thus the close time is known by the microsynchronizer from a pattern of wires on the socket where the computer card plugs in. In order to protect against the chance that one of these wires would short or open, the pattern is actually repeated on four more pins, but is reversed (one's complement). The computer reads the close time twice; once from each pattern. It compares the close times and proceeds only if they agree.

Another feature is that the close command is withdrawn when the predicted phase $\emptyset_F$ and slip $\dot{\emptyset}_F$ leave the target area. This provides protection should the breaker fail to move when commanded.

Associated with this, the relay used in many synchronizers is incapable of interrupting the breaker's closing solenoid current. Normally, powerful auxiliary contacts on the breaker itself interrupt the close coil. However, if the breaker fails to close, the synchronizer relay must interrupt this highly inductive DC load. The relay chosen for the Microsynchronizer control is capable of doing so with no suppression components.

Another safety precaution is that both a "one" and a "zero" output are required from the same microcomputer output port to effect closure of the relay on the isolation module. The relay has one transistor above the coil and one below. Both must turn on to make the relay close. In this way, both output channels must be active and most failure modes of the circuit component will not cause a false close command.

Although the Microsynchronizer control is designed to work with or without voltage matching, it does not perform that function. If voltage matching is required, a separate device is used. The synchronizer does make a check to see if reasonable signals are present from the line and generator potential transformers. The line voltage must be above 50% of its nominal value (this permits emergency closures onto a system that needs the added generation capacity of the gas turbine generator) and the generator voltage above 85% (this allows closure onto a system whose voltage is low and it is desired to go on-line with the lowest possible setting of the generator output voltage) for the microsynchronizer to take action. Thus, if a potential transformer fuse is blown, or if the generator excitation system has failed low, the microsynchronizer will be disabled.

One of the requirements for good synchronizer performance is proper preparation and filtration of the incoming signals. Careful attention was given to the suppression of line transients and common mode signals such as those introduced by lightning or ground potential rise. The isolation module uses special shielded transformers, metal oxide varistors, and junction transistor circuits to pass the signal and suppress the transients. Further filtration is accomplished, in the computer itself, using digital techniques.

TROUBLE SHOOTING

Trouble shooting is performed by replacing the module or the card. Refer to Figure 6. If a system is not functional, perform the following checks:

a. Verify that P28V exists on module TB2-2, TB2-3.

b. Line and generator voltages must be present. Line voltage must exceed 65Vrms, and generator voltage must exceed 80Vrms. These conditions are indicated by two Module Light Emitting Diodes (LEDs); ON, meaning voltage present. Verify that both LEDs are on.

c. If the two LEDs are on, square 60 Hz pulses should be generated on TB2-9 and TB2-11 with respect to TB2-10 and TB2-12. Refer to Figure 6. A DC voltmeter should read approximately 2.5 VDC each. An oscilloscope should display a square wave from 0 to 5 VDC, 60 Hz on each. If not, replace the isolating module.

d. If the LEDs will not come on and the generator and line voltages are present, verify that 5 volts is present on TB2-8 with respect to TB2-10. Replace the isolation module if voltage is present. If P5V is not present, check the wiring to the module.

e. If the computer card "RAISE", "LOWER" is non functional, verify the following inputs:

- card pin 38 (ORST) = logic 1

- card pin 30 (OSTRT) = logic 0

- card pin 23 with no connection

- pulses as defined in part C on pins 11 and 9 with respect to pins 12 and 10.

Replace the computer card if all the above connections and conditions are true.

f. If the RAISE, LOWER signals are functional, but the breaker will not close, verify that:

- card pin 32 (OVMTCH) is logic 0

- card pin 17 pulses to logic 1

- card pin 16 pulses to logic 0

The pulses on pins 16 and 17 should occur when the synchroscope is close to 0 degree phase shift. If pulses do not occur, replace the computer card. If pulses are present, verify the relay contacts in the isolation module are closing. If relay contact is not closing, replace the isolation module. If the relay contact is closing, verify correct wiring and then the breaker closing mechanism.

HARDWARE DEFINITION

Isolation module IC3602A178

Computer card IC3600SASC

For wiring information, refer to drawing information on the 3 series of sheets for respective device IC3602A178 and IC3600SASC.

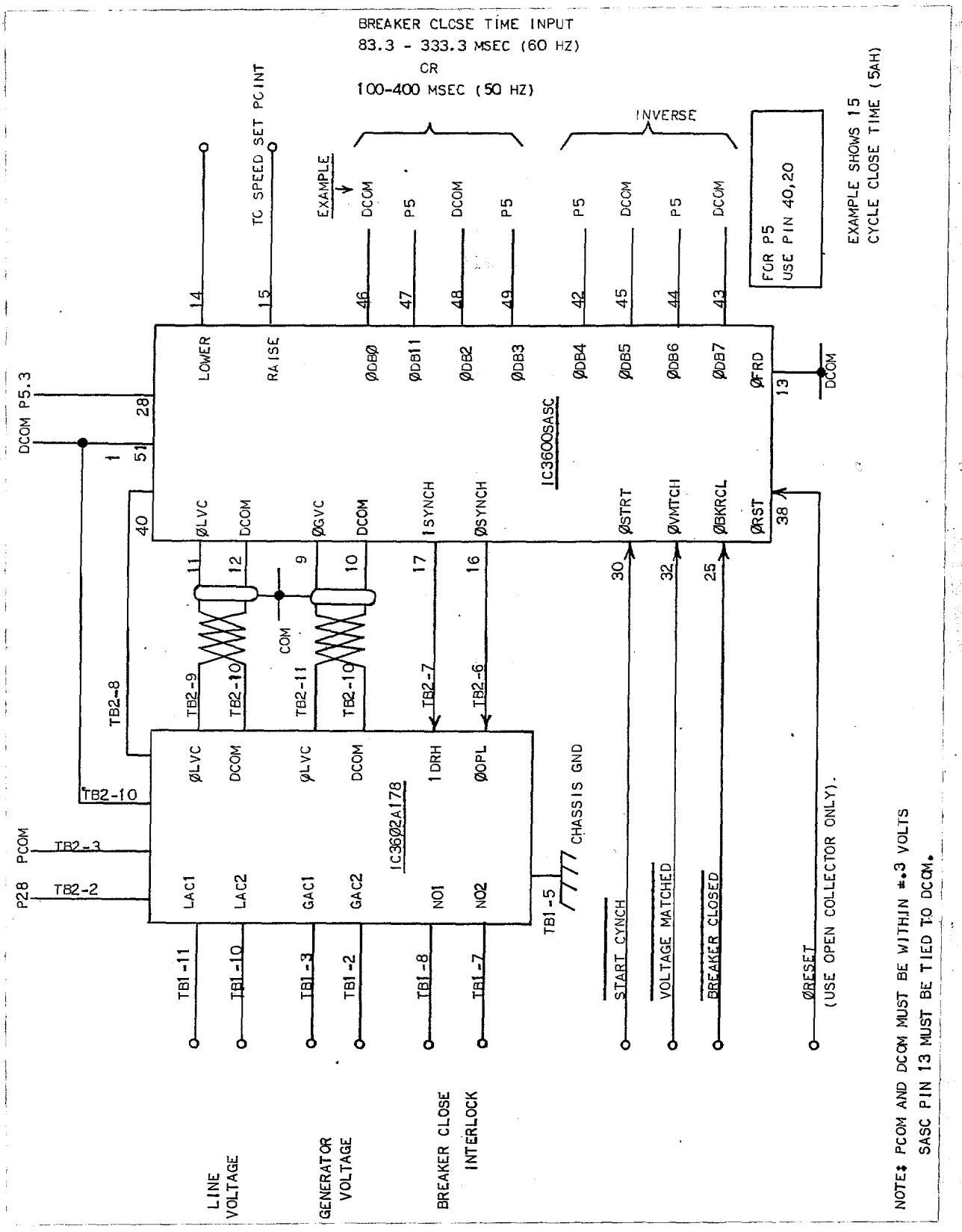

BREAKER CLOSE TIME INPUT
83.3 - 333.3 MSEC (60 HZ)
OR
100-400 MSEC (50 HZ)
TC SPEED SET POINT
EXAMPLE
INVERSE
FOR P5 USE PIN 40,20
EXAMPLE SHOWS 15 CYCLE CLOSE TIME (5AH)
DCOM P5.3
IC3600SASC
LOWER
RAISE
ØDBØ
ØDB11
ØDB2
ØDB3
ØDB4
ØDB5
ØDB6
ØDB7
ØFRD
ØLVC
DCOM
ØGVC
ISYNCH
ØSYNCH
ØSTRT
ØVMTCH
ØBKRCL
ØRST
COM
TB2-8
TB2-10
PCOM
P28
TB2-3
TB2-2
IC3602A178
LAC1
LAC2
GAC1
GAC2
NO1
NO2
IDRH
ØOPL
CHASSIS GND
TB1-5
TB1-11
TB1-10
TB1-3
TB1-2
TB1-8
TB1-7
START CYNCH
VOLTAGE MATCHED
BREAKER CLOSED
ØRESET
(USE OPEN COLLECTOR ONLY).
LINE VOLTAGE
GENERATOR VOLTAGE
BREAKER CLOSE INTERLOCK
NOTE: PCOM AND DCOM MUST BE WITHIN ±.3 VOLTS
SASC PIN 13 MUST BE TIED TO DCOM.

Figure 6

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